

Technical Documentation: Football Data Pipeline

Data Lake House Implementation on AWS for Advanced Analytics

Project developed for

Federación Mexicana de Fútbol (FMF)

Implemented by

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1. Environment Configuration

This document describes the infrastructure components, automation scripts, and logical data model design implemented on AWS for the processing and analysis of historical match data from national football teams. The solution processes data from an external API and consolidates it into an optimized columnar store in Amazon S3, governed by AWS Glue and queried through Amazon Athena.

2. Infrastructure Resource Inventory

AWS Service	Name/Identifier	Primary Function
Amazon EC2	unam-2026-ingenieriaded atos-grupo6	Instance with the necessary setup to run all Python scripts required for the project
Amazon S3	unam-2026-ingenieriada tos-equipo6	Centralized Data Lake bucket storing the three data maturity layers (/1bronze/ , /2silver/ , /3gold/)
AWS Glue Job	silver1 (Engine: Spark 3.3 , Python 3)	Serverless ETL Job that processes the PySpark script for data cleansing, typing, and partitioning.
AWS Glue Crawler	crawler-silver-parquet	Crawler responsible for inspecting partitioned folders in S3, inferring the schema, and registering metadata

AWS Service	Name/Identifier	Primary Function
Amazon Athena	Motor Serveless (Query Engine V3)	Environment for executing distributed analytical queries in standard SQL over the data catalog.

3. Pipeline Specifications (ETL)

3.1. Ingestion (API-Football ——— S3 Bronze Layer)

Project data ingestion is performed using Python scripts executed on an Amazon EC2 instance. These scripts connect to API-Football using HTTP requests authenticated with an API Key to retrieve historical data on international matches from various national teams.

The process begins by querying multiple seasons per selected team, retrieving data such as match date, home team, away team, goals scored, and corresponding tournament. The data obtained in JSON format is then transformed and filtered to retain only matches played since August 2022.

Once processed, the records are stored in individual CSV files per national team. Then, through another automated script, these files are uploaded to Amazon S3 within the Bronze layer of the Data Lake, organized in folders by country.

3.2. Transformation (EMR/Glue ——— Silver)

The AWS Glue job runs a distributed PySpark script (silver1.py) that strictly applies the following data transformations and quality rules:

- *Correct encoding:*

Forced UTF-8 source reading to prevent character encoding issues with accented characters in team names

- *String cleansing:*

Application of .WithColumn() functions combined with upper() and trim() to standardize rival team names and eliminate residual whitespace

- *Type casting:*

Conversion of text columns to native data types suitable for numerical analytics:

- Goles_local — IntegerType()
- Goles_visitante — IntegerType()
- Fecha — TimestampType()

- *Range filtering as a quality rule:*

Elimination of anomalies using filter() clauses, allowing only records where the goal count falls within a logical range of 0 to 20 per match.

- *Partitioning optimization:*

The clean DataFrame is written to the Silver Layer using the optimized columnar Parquet format, applying partitioning by the year control column.

- Output path:

s3://unam-2026-ingenieriadedatos-equiposilver/[seleccion]/year=YYYY/

3.3. Refinement (S3 Silver Layer — Amazon Athena — Gold Layer)

An AWS Glue Crawler dynamically maps the root of the Silver layer. Upon identifying the subfolders generated by PySpark, it updates the partitions within the Data Catalog, exposing the partitioned table to Amazon Athena.

Through Amazon Athena, analytical queries are executed to generate aggregated views with key KPIs (total matches, wins, draws, losses, win rate %, goals scored and conceded). The high business-value results are stored in the final layer at the path:

s3://unam-2026-ingenieriadedatos-equipogold/

4. Data Dictionary (Gold Layer)

A schema of the resulting data types in the catalog **glue-2silver.2silver**:

Name	Data Type (Athena)	Business Rule
<i>fecha</i>	timestamp	Date and time when the match was played
<i>torneo</i>	string	Official name of the competition
<i>equipo_local</i>	string	Name of the team that played as the home side (In uppercase)
<i>equipo_visitante</i>	string	Name of the team that played as the away side (In uppercase)
<i>goles_local</i>	int	Number of goals scored by the home team (0-20)
<i>goles_visitante</i>	int	Number of goals scored by the away team (0-20)
<i>partition_0</i>	string	Primary partition key corresponding to the analyzed country
<i>year</i>	string	Secondary partition key (Match year)

5. Execution and Monitoring Guide

To manually start the distributed ETL processing using the AWS command-line interface, run the following instruction:

```
aws glue start-job-run --job-name silver1
```

The execution status is audited directly in the AWS Glue Monitoring console and print streams are directed to specific logs in Amazon CloudWatch.

Performance tests run on the Serverless cluster show an average processing time of 1 minute and 54 seconds for the bulk transformation from the Bronze to Silver layer, making it highly efficient and scalable as data volumes grow.